

Model Formulation 3

Sets

- N : set of physical nodes
- A : set of physical arcs
- D : set of demands
- F : set of VNF types
- τ : virtual sink node ($\tau \notin N$)
- $N^+ = N \cup \{\tau\}$: extended node set
- $A^+ = A \cup \{(i, \tau) : i \in N\}$: extended arc set (virtual arcs have unlimited capacity)
- $O = \{o^k : k \in D\}$: set of origin nodes

Parameters

- u : capacity of each physical arc $(i, j) \in A$
- q_f : capacity of service type $f \in F$
- d^k : bandwidth demand of $k \in D$
- o^k : origin node of demand $k \in D$ ($o^k \in N$)
- n : number of services in the chain (same for all demands)
- $f : \{1, \dots, n\} \rightarrow F$: type of service at position s in the chain
- $\alpha_{i,f} \in \{0, 1\}$: equals 1 if service type f can be executed on node i and $i \notin O$, 0 otherwise
- $\tau_{ij} \geq 0$: transport delay on physical arc $(i, j) \in A$ (virtual arcs have zero delay)
- $\sigma_f \geq 0$: fixed setup time for VNF type $f \in F$
- $\gamma_f \geq 0$: variable processing time per demand for VNF type $f \in F$

Decision Variables

- $y_{if} \in \{0, 1\}$: 1 if service type $f \in F$ is installed on node $i \in N$
- $z_{if}^k \in \{0, 1\}$: 1 if demand $k \in D$ uses service type $f \in F$ on node $i \in N$
- $x_{ij}^{ks} \in \{0, 1\}$: 1 if extended arc $(i, j) \in A^+$ is used by demand $k \in D$ in subpath $s \in \{1, \dots, n+1\}$
- $w_{uv}^{i,s+1} \in \{0, 1\}$: 1 if extended arc $(u, v) \in A^+$ is used in subpath $s+1$ by the aggregated flow that has just completed service s on node i ($i \in N$, $s \in \{1, \dots, n\}$)
- $T_{if} \in \mathbb{R}_+$: total processing time at node $i \in N$ for VNF type $f \in F$
- $\Theta_{if}^k \in \mathbb{R}_+$: processing time experienced by demand $k \in D$ when using VNF type $f \in F$ at node $i \in N$
- $L^k \in \mathbb{R}_+$: total end-to-end latency for demand $k \in D$ (transport + processing)
- $L_{\max} \in \mathbb{R}_+$: maximum latency among all demands

Objective Function

$$\min L_{\max} \tag{1}$$

Constraints

$$\sum_{i \in N} z_{if}^k = 1, \quad \forall k \in D, f \in F \quad (2)$$

$$z_{if}^k \leq y_{if}, \quad \forall k \in D, i \in N, f \in F : \alpha_{i,f} = 1 \quad (3)$$

$$z_{if}^k = 0, \quad \forall k \in D, i \in N, f \in F : \alpha_{i,f} = 0 \quad (4)$$

$$y_{if} = 0, \quad \forall i \in N, f \in F : \alpha_{i,f} = 0 \quad (5)$$

$$\sum_{k \in D} z_{if}^k \leq q_f, \quad \forall i \in N, f \in F : \alpha_{i,f} = 1 \quad (6)$$

$$\sum_{j:(i,j) \in A^+} x_{ij}^{ks} - \sum_{j:(j,i) \in A} x_{ji}^{ks} = z_{i,f(s-1)}^k - z_{i,f(s)}^k, \quad \forall k \in D, i \in N, s \in \{2, \dots, n\} \quad (7)$$

$$\sum_{j:(i,j) \in A} x_{ij}^{k1} - \sum_{j:(j,i) \in A} x_{ji}^{k1} = \begin{cases} 1 - z_{i,f(1)}^k, & i = o^k \\ -z_{i,f(1)}^k, & i \neq o^k \end{cases}, \quad \forall k \in D, i \in N \quad (8)$$

$$\sum_{j:(i,j) \in A^+} x_{ij}^{k,n+1} - \sum_{j:(j,i) \in A} x_{ji}^{k,n+1} = z_{i,f(n)}^k, \quad \forall k \in D, i \in N \quad (9)$$

$$\sum_{s=1}^{n+1} \sum_{l:(j,i) \in A} x_{ji}^{ks} \leq 1, \quad \forall k \in D, i \in N \quad (10)$$

$$\sum_{s=1}^{n+1} \sum_{j:(i,j) \in A^+} x_{ij}^{ks} \leq 1, \quad \forall k \in D, i \in N \quad (11)$$

$$x_{uv}^{k,s+1} \leq w_{uv}^{i,s+1} + (1 - z_{i,f(s)}^k), \quad \forall k \in D, i \in N, (u,v) \in A^+, s \in \{1, \dots, n\} \quad (12)$$

$$\sum_{v:(u,v) \in A^+} w_{uv}^{i,s+1} \leq 1, \quad \forall u \in N, i \in N, s \in \{1, \dots, n\} \quad (13)$$

$$T_{if} = \sigma_f \cdot y_{if} + \gamma_f \cdot \sum_{k \in D} z_{if}^k, \quad \forall i \in N, f \in F : \alpha_{i,f} = 1 \quad (14)$$

$$\Theta_{if}^k \leq M \cdot z_{if}^k, \quad \forall k \in D, i \in N, f \in F : \alpha_{i,f} = 1 \quad (15)$$

$$\Theta_{if}^k \geq T_{if} - M(1 - z_{if}^k), \quad \forall k \in D, i \in N, f \in F : \alpha_{i,f} = 1 \quad (16)$$

$$L^k = \sum_{s=1}^{n+1} \sum_{(i,j) \in A} \tau_{ij} \cdot x_{ij}^{ks} + \sum_{\substack{i \in N, f \in F \\ \alpha_{i,f}=1}} \Theta_{if}^k, \quad \forall k \in D \quad (17)$$

$$L_{\max} \geq L^k, \quad \forall k \in D \quad (18)$$

Explanation of Constraints

(2) **VNF assignment:** each demand must be served by exactly one instance of each required VNF type.

(3)–(5) **Feasibility and installation:** a VNF can be used on a node only if it is installed there (3), and both usage (4) and installation (5) are forbidden on nodes where that VNF type is not executable ($\alpha_{i,f} = 0$).

- (6) **VNF capacity:** the total bandwidth processed by service type f on node i cannot exceed q_f . Multiplying by $\alpha_{i,f}$ makes the right-hand side zero when f is not executable on i .
- (7) **Intermediate flow balance:** for subpaths $s = 2, \dots, n$, the net flow entering node i equals the difference between using service $f(s-1)$ and service $f(s)$ at i . This links successive services along the chain.
- (8) **First subpath flow balance:** from the origin o^k , one unit of flow is injected; at other nodes the net outflow equals $z_{i,f(1)}^k$, forcing the first VNF to be reached exactly once.
- (9) **Last subpath flow balance:** after the final service $f(n)$, the flow is directed to the virtual sink τ through arcs $(i, \tau) \in A^+$. The net flow at node i equals $z_{i,f(n)}^k$, meaning the flow leaves the node where the last VNF was processed.
- (10)–(11) **Simple path:** each demand uses at most one incoming and one outgoing arc at any node, preventing cycles and traffic splitting.
- (12) **Path unification:** if demand k uses service s on node i ($z_{i,f(s)}^k = 1$), then its subsequent routing ($x_{uv}^{k,s+1}$) must follow the aggregated path $w_{uv}^{i,s+1}$ that originates from i . This forces all demands sharing the same service node to travel together toward the sink.
- (13) **No bifurcation of aggregated paths:** the aggregated path leaving node i after service s cannot branch; at most one outgoing arc per node is used. Together with (12), this guarantees a single tree rooted at τ for each service-stage.
- (14) **Processing time:** the total processing time at node i for VNF f includes a fixed setup time σ_f (paid once if $y_{if} = 1$) plus a variable time γ_f per demand using that VNF.
- (15)–(16) **Linearization:** $\Theta_{if}^k = T_{if}$ if demand k uses VNF f at node i ($z_{if}^k = 1$), otherwise $\Theta_{if}^k = 0$. Here M is a sufficiently large constant.
- (17) **End-to-end latency:** total time for demand k is the sum of transport delays on all arcs used plus processing times at all VNFs visited
- (18) **Maximum latency:** $L_{\max}$ is forced to be at least as large as the latency of every demand. Minimizing $L_{\max}$ in the objective ensures it equals the maximum latency across all demands..

Note on the big-M constant: The constant M in constraints (15)–(16) must be chosen large enough to ensure that Θ_{if}^k can take the value of T_{if} when $z_{if}^k = 1$. A valid choice is:

$$M = \max_{f \in F} \sigma_f + \max_{f \in F} \gamma_f \cdot |D|$$

This bound is sufficient because the maximum processing time T_{if} occurs when all demands use the same VNF instance, giving $T_{if} \leq \sigma_f + \gamma_f \cdot |D| \leq M$.